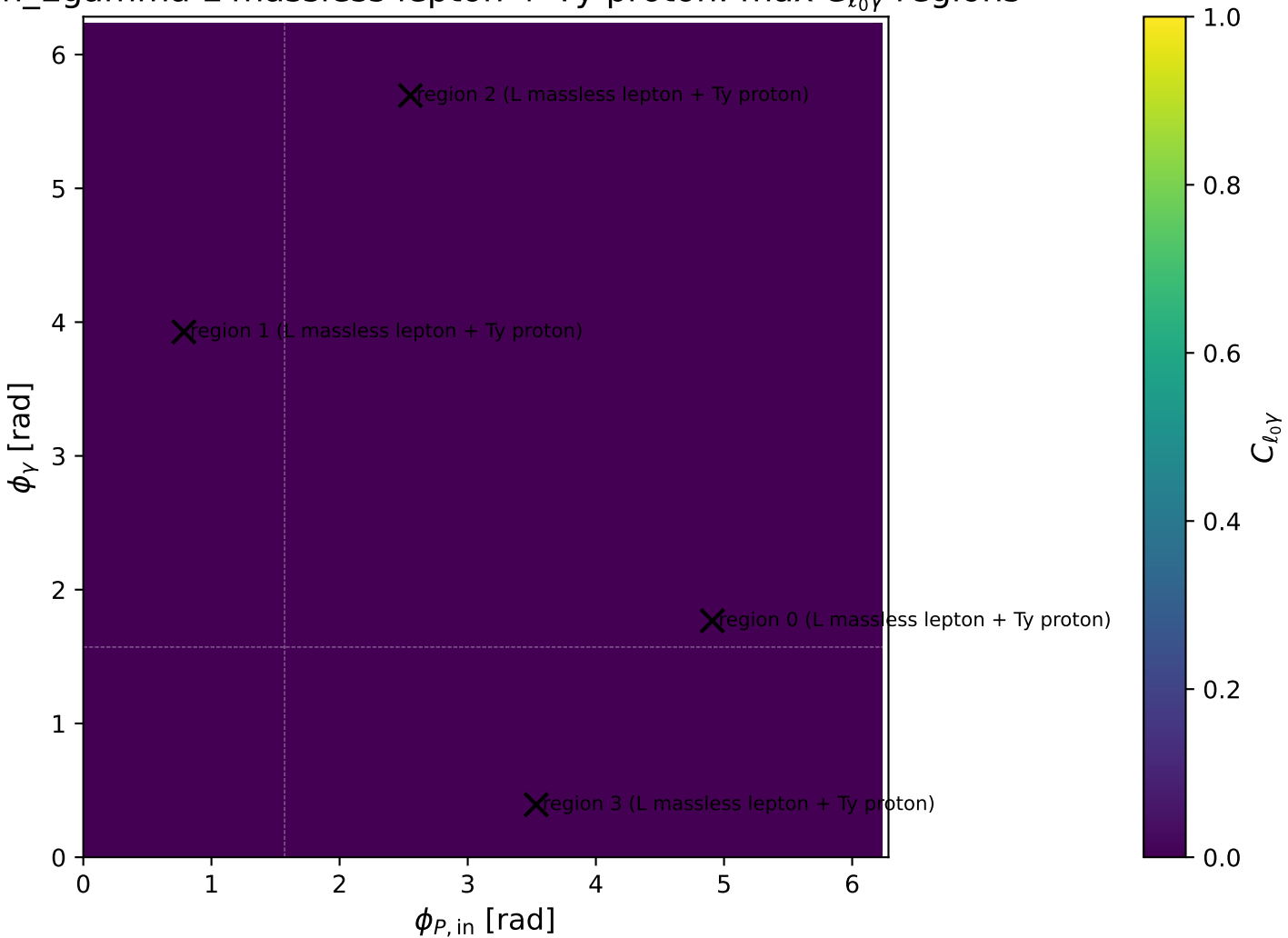
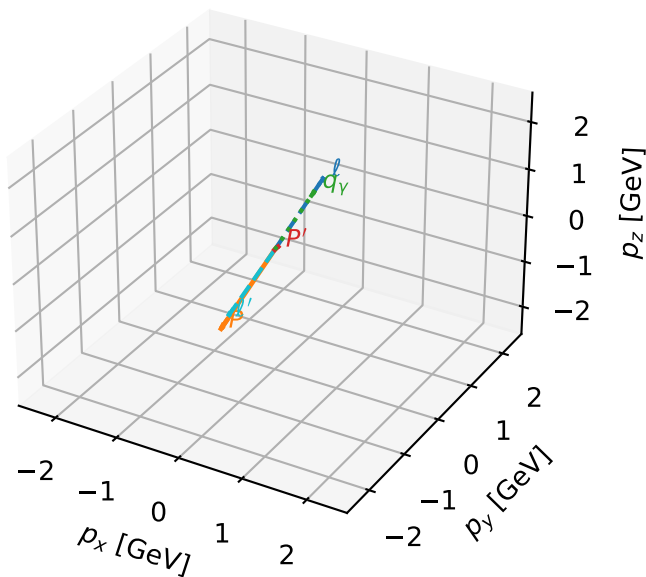


# h\_Gamma L massless lepton + Ty proton: max $C_{l_0\gamma}$ regions



# L massless lepton + Ty proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta



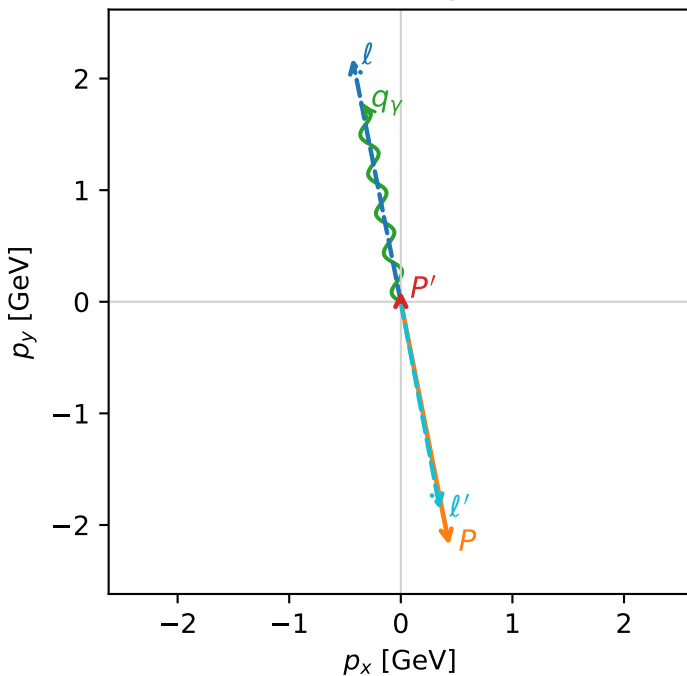
c\_e\_gamma\_high\_egamma\_lty\_region\_0  $C_{\ell_0\gamma}=2.55533\text{e-}11$   
 region: high\_Egamma\_LTy\_region\_0 final-pair delta\_xy=3.13158 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_{t_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.0972622$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.76715$ ,  $\phi_p=4.90874$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=1.76715$   
 $Q^2=16.865$ ,  $x_B=0.846786$ ,  $t=-3.21783$ ,  $W^2=3.93133$ ,  $y=0.965134$   
 $\theta(\ell', \gamma)=179.426$  deg

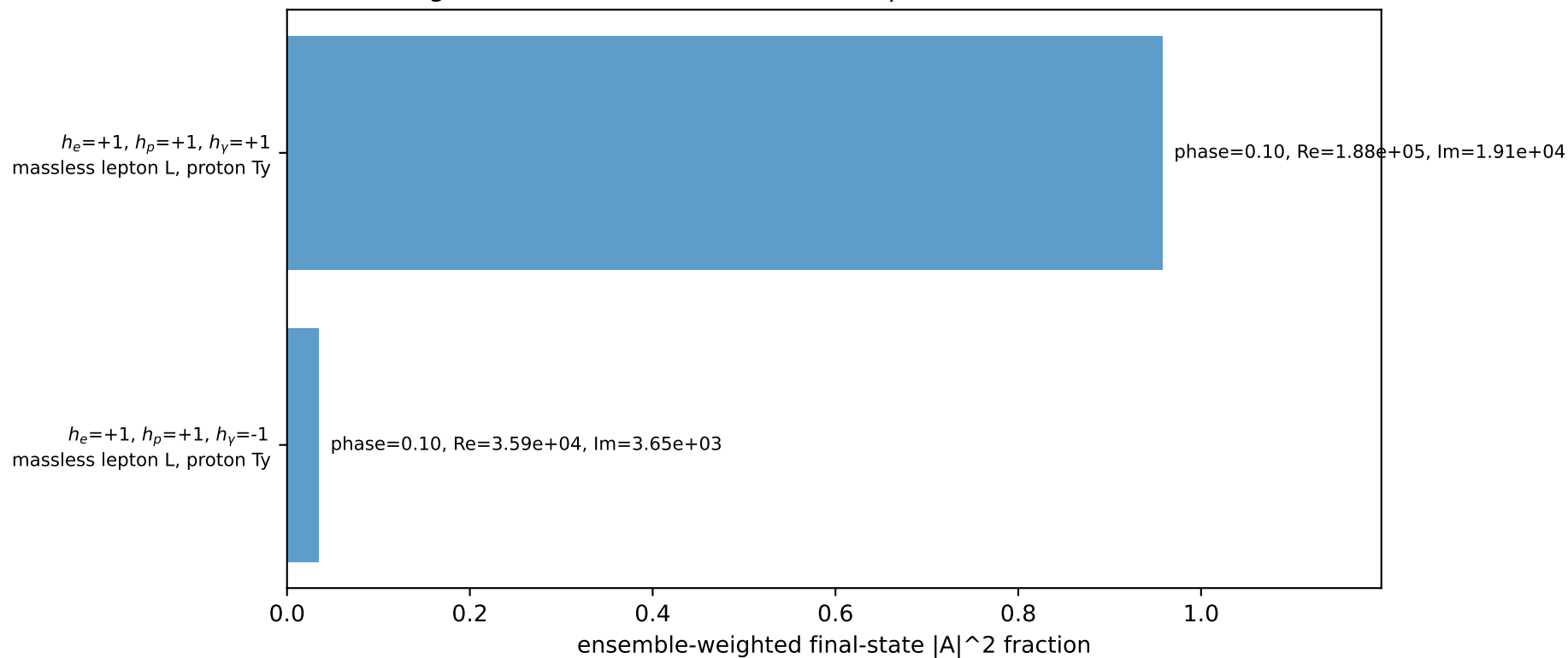
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |             |                |                |                |
|------------|-------------|----------------|----------------|----------------|
| $\ell$     | $E= 2.2244$ | $p_x= -0.4340$ | $p_y= 2.1817$  | $p_z= -0.0000$ |
| $P$        | $E= 2.4141$ | $p_x= 0.4340$  | $p_y= -2.1817$ | $p_z= 0.0000$  |
| $\ell'$    | $E= 1.8955$ | $p_x= 0.3512$  | $p_y= -1.8627$ | $p_z= 0.0000$  |
| $P'$       | $E= 0.9430$ | $p_x= 0.0000$  | $p_y= 0.0973$  | $p_z= 0.0000$  |
| $q_\gamma$ | $E= 1.8000$ | $p_x= -0.3512$ | $p_y= 1.7654$  | $p_z= 0.0000$  |

Transverse plane

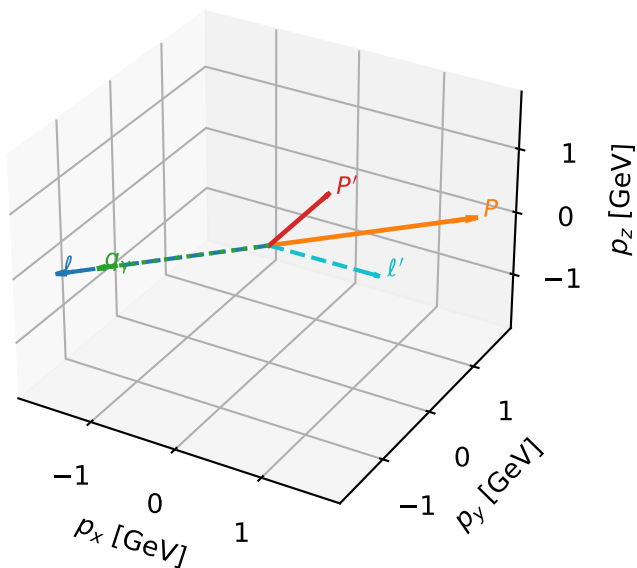


Leading initial-ensemble/final-state components (N=2, retained=99.3%)



# L massless lepton + Ty proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta



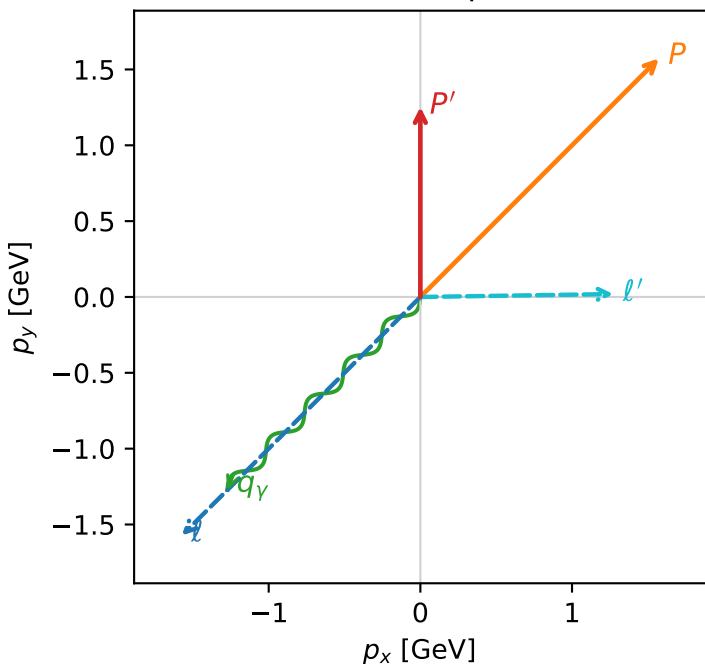
c\_e\_gamma\_high\_egamma\_lty\_region\_1 C\_{\ell\_0\gamma}=1.99733e-11  
region: high\_Egamma\_LTy\_region\_1 final-pair delta\_xy=2.37137 rad  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $|L_{\ell_0}\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.25347$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.92699$ ,  $\phi_p=0.785398$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=3.92699$   
 $Q^2=9.72782$ ,  $x_B=0.524277$ ,  $t=-1.85606$ ,  $W^2=9.70675$ ,  $y=0.899144$   
 $\theta(\ell', \gamma)=135.87$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |      |        |        |         |        |         |        |         |
|------------|------|--------|--------|---------|--------|---------|--------|---------|
| $\ell$     | $E=$ | 2.2244 | $p_x=$ | -1.5729 | $p_y=$ | -1.5729 | $p_z=$ | -0.0000 |
| $P$        | $E=$ | 2.4141 | $p_x=$ | 1.5729  | $p_y=$ | 1.5729  | $p_z=$ | 0.0000  |
| $\ell'$    | $E=$ | 1.2729 | $p_x=$ | 1.2728  | $p_y=$ | 0.0193  | $p_z=$ | 0.0000  |
| $P'$       | $E=$ | 1.5656 | $p_x=$ | 0.0000  | $p_y=$ | 1.2535  | $p_z=$ | 0.0000  |
| $q_\gamma$ | $E=$ | 1.8000 | $p_x=$ | -1.2728 | $p_y=$ | -1.2728 | $p_z=$ | 0.0000  |

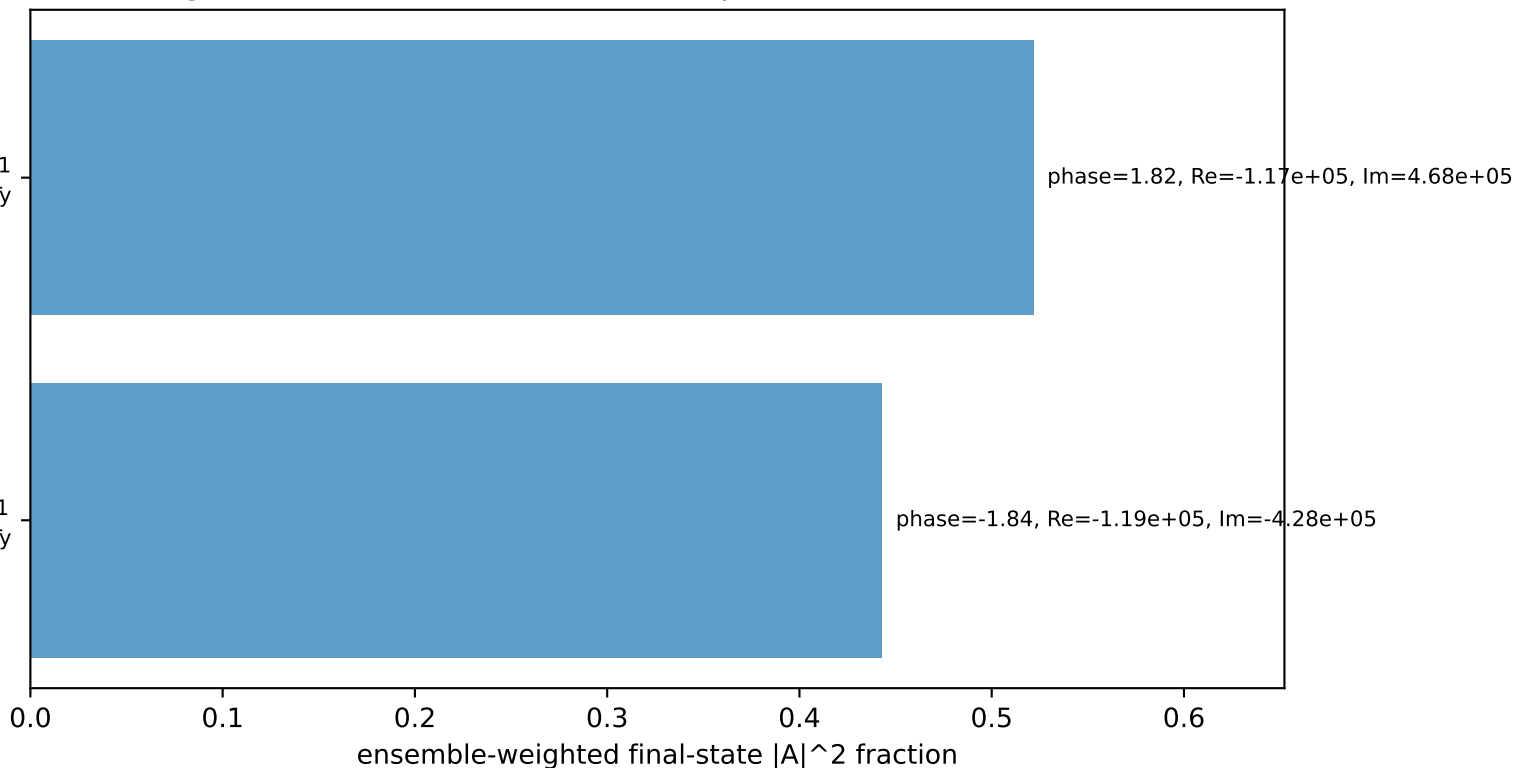
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
massless lepton L, proton Ty

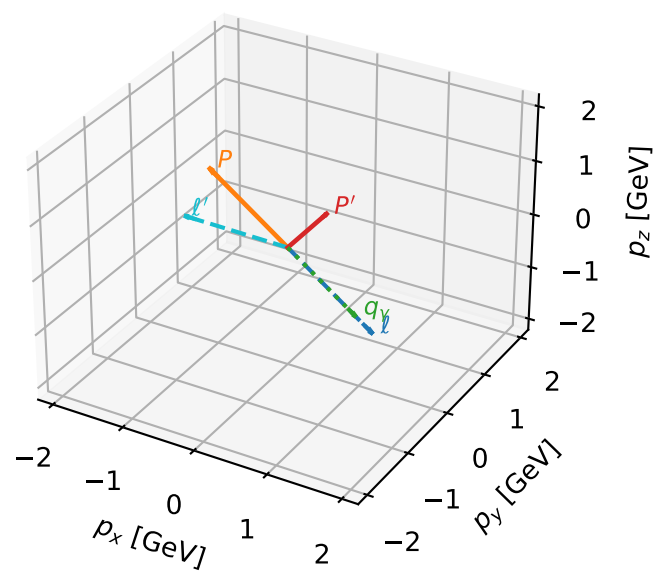
$h_e=+1, h_p=-1, h_\gamma=+1$   
massless lepton L, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=96.5%)



# L massless lepton + Ty proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta

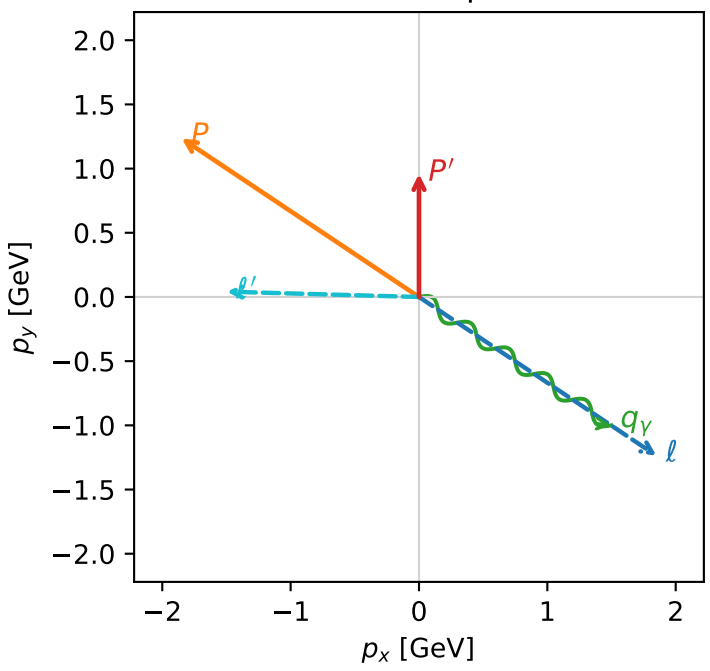


c\_e\_gamma\_high\_egamma\_lty\_region\_2 C\_{l\_0\gamma}=1.50457e-11  
region: high\_Egamma\_LTy\_region\_2 final-pair delta\_xy=2.5801 rad  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

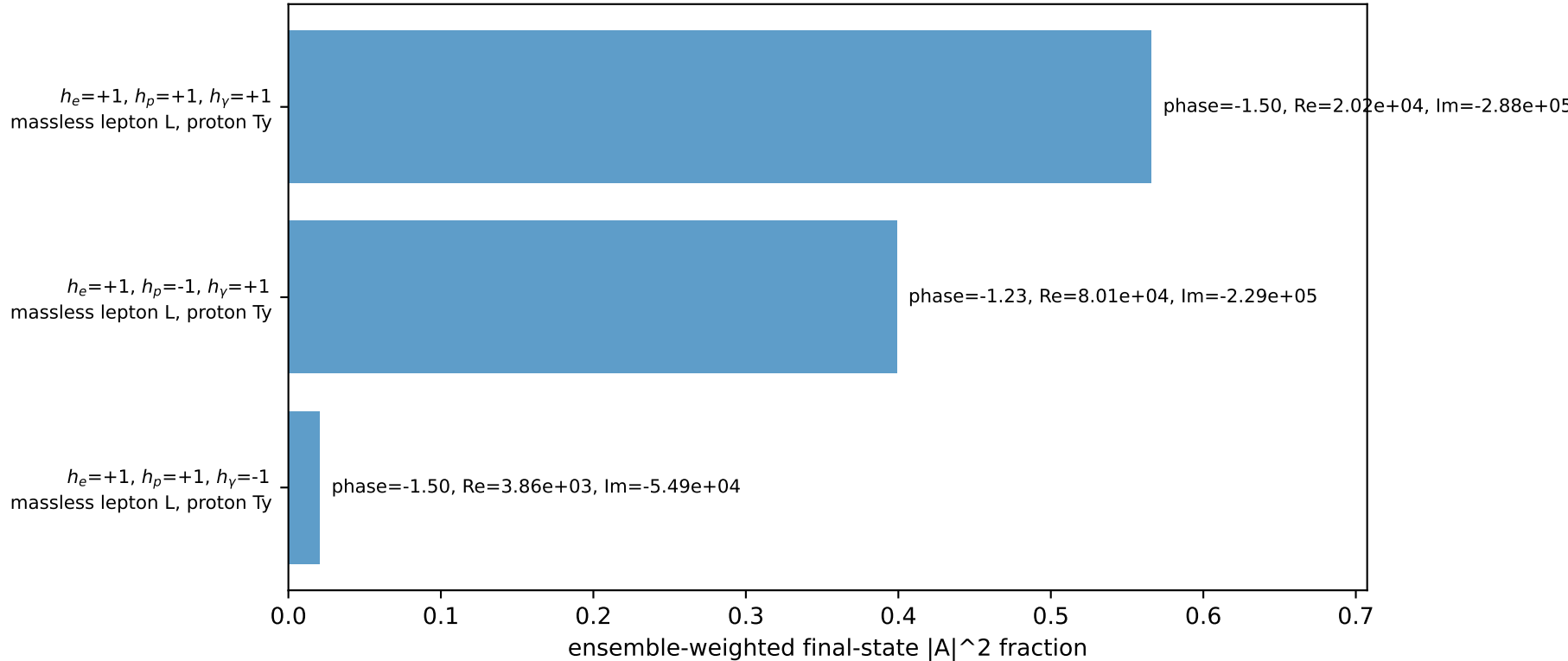
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.958776$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=5.69414$ ,  $\phi_P=2.55254$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=5.69414$   
 $Q^2=12.299$ ,  $x_B=0.645776$ ,  $t=-2.34664$ ,  $W^2=7.62614$ ,  $y=0.922917$   
 $\theta(\ell', \gamma)=147.829$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 1.8495$   $p_y= -1.2358$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -1.8495$   $p_y= 1.2358$   $p_z= 0.0000$   
 $\ell'$   $E= 1.4972$   $p_x= -1.4966$   $p_y= 0.0413$   $p_z= 0.0000$   
 $P'$   $E= 1.3413$   $p_x= 0.0000$   $p_y= 0.9588$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 1.4966$   $p_y= -1.0000$   $p_z= 0.0000$

Transverse plane

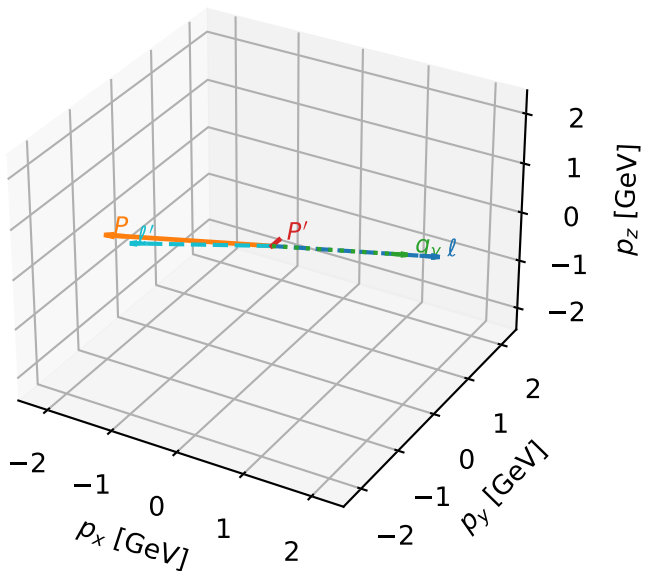


Leading initial-ensemble/final-state components (N=3, retained=98.5%)



# L massless lepton + Ty proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta

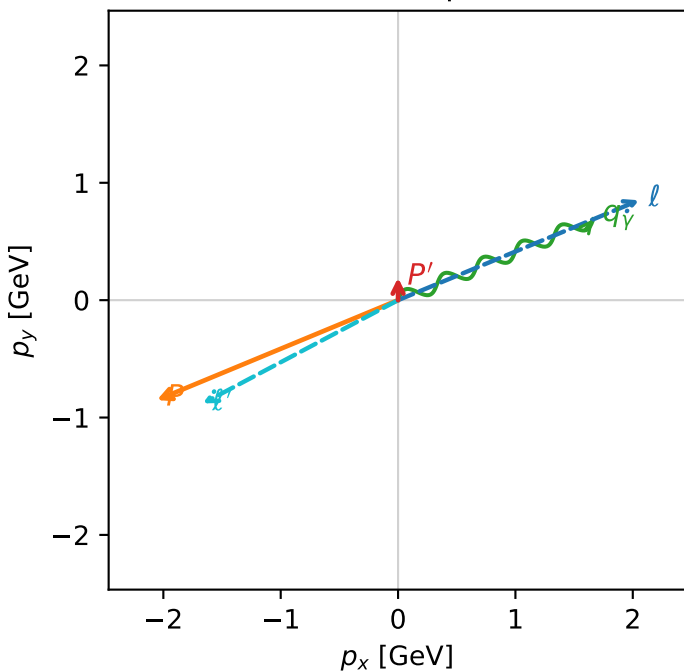


c\_e\_gamma\_high\_egamma\_lty\_region\_3  $C_{\ell_0\gamma}=9.66221e-12$   
 region: high\_Egamma\_LTy\_region\_3 final-pair delta\_xy=3.04774 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.190824$   
 $\theta_{in}=1.57079$ ,  $\phi_{\ell}=0.392699$ ,  $\phi_P=3.53429$   
 $E_{\gamma}=1.8$ ,  $\phi_{\gamma}=0.392699$   
 $Q^2=16.7024$ ,  $x_B=0.839929$ ,  $t=-3.18681$ ,  $W^2=4.06292$ ,  $y=0.96363$   
 $\theta(\ell', \gamma)=174.623$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 2.0551$   $p_y= 0.8512$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.0551$   $p_y= -0.8512$   $p_z= 0.0000$   
 $\ell'$   $E= 1.8813$   $p_x= -1.6630$   $p_y= -0.8797$   $p_z= 0.0000$   
 $P'$   $E= 0.9572$   $p_x= 0.0000$   $p_y= 0.1908$   $p_z= 0.0000$   
 $q_{\gamma}$   $E= 1.8000$   $p_x= 1.6630$   $p_y= 0.6888$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.2%)

